Category Theory for Mathematical Physics — Exercise Sheet 11

Algebras in tensor categories and module categories

Variant: Hints and answers

Conventions as in Lecture 11: C is a fusion category, $\mathcal{M}$ a left C-module category, $[M,N]$ the internal Hom (Definition 11.7); ψ is a 2-cochain (in Exercise 6, ψ is Ising's fermion instead), and $(\delta\psi)$, ω are as in Lecture 3. The *NIM-rep* of a semisimple module category $\mathcal{M}$ with simple objects $M_1,\dots,M_r$ is the family of matrices $(n_a)_{ij} = \dim \mathrm{Hom}_{\mathcal{M}}(a \otimes M_j, M_i)$, one for each simple a of C. In general, a *NIM-rep* of the fusion ring of C is a family of square non-negative integer matrices n_a, all of one size $r \geq 1$, with $n_1 = 1$ for the unit (n_0 in Exercise 5), $n_a n_b = \sum_c N_{ab}^c\, n_c$ and $n_{a^*} = n_a^T$; it is *indecomposable* if no nonempty proper set of indices (simple objects) is closed under every n_a. Module categories on this sheet are nonzero. $\mathbb{Z}_N$ is additive, $\mathbb{Z}_2 = \{1, s\}$ and S_3 multiplicative. In Exercise 2, ε is Sheet 3's carry (not the toric fermion) and $\zeta_4 = i$. In Exercise 5, U_m is the polynomial with $U_0 = 1$, $U_1 = x$, $U_{m+1} = xU_m - U_{m-1}$, so that $U_m(2\cos y) = \sin((m+1)y)/\sin y$; in Exercise 6, π is an involution of the simple objects.

Exercise 1 (Module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$). (i) Using Theorem 11.11, list the indecomposable semisimple module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$, with their simple objects and the matrix n_s.

(ii) The direct sum $\mathrm{Vec}_{\mathbb{Z}_2} \oplus \mathrm{Vect}_{\mathrm{fd}}$ is a module category too. Write its n_s, and say why it is not on the list.

Hint. (i) Two subgroups, and $H^2(\mathbb{Z}_2, \mathbb{C}^\times) = 0$ (Example 11.12).

Answer. (i) $H = 1$: the regular module, simples δ_1, δ_s, $n_s = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; $H = \mathbb{Z}_2$: $\mathrm{Vect}_{\mathrm{fd}}$, one simple, $n_s = (1)$. (ii) $n_s = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, block diagonal: the module category is decomposable.

Exercise 2 (The algebra of a rank-one module). Let ω_p be the cocycle of Sheet 3 on $\mathbb{Z}_4$ and $H = \{0, 2\}$.

(i) Write the module pentagon of Theorem 11.6 at $(g,h,k) = (2,2,2)$ for a normalised ψ, and show that a rank-one module category over the part of $\mathrm{Vec}_{\mathbb{Z}_4}^{\omega_p}$ on H exists exactly when p is even.

(ii) For such a module $\mathcal{M} = \mathrm{Vect}_{\mathrm{fd}}$, with $M = \mathbb{C}$, compute $[M,M]$ by Theorem 11.8 and its product on the basis $e_h \in \mathrm{Hom}(\delta_h \otimes M, M)$, and show that it is the twisted group algebra $\mathbb{C}_\psi[H]$ of Example 11.2.

(iii) For $\mathrm{Mod}(\mathbb{C}[H])$ over Vec_G (trivial ω, $H \leq G$), show that $[M_H, M_H] \cong \mathbb{C}[H]$ as algebras.

Hint. (i) $\psi(0,2) = \psi(2,0) = 1$ and $2+2 = 0$. (ii) The product is the morphism corresponding to $\mathrm{ev} \circ (\mathrm{id} \otimes \mathrm{ev}) \circ \alpha^{\mathcal{M}}$. (iii) $\mathrm{Hom}_A(\delta_g \otimes A, A) \cong \mathrm{Hom}(\delta_g, A)$.

Answer. (i) $\psi(2,2)\psi(0,2) = \omega_p(2,2,2)\psi(2,2)\psi(2,0)$, i.e. $1 = \omega_p(2,2,2) = (-1)^p$. (ii) $[M,M] = \delta_0 \oplus \delta_2$, and $e_g e_h = \psi(g,h)e_{g+h}$: associative exactly by the module pentagon. (iii) $[M_H, M_H] = \bigoplus_{h\in H} \delta_h$, with e_g left multiplication by g: $e_g e_h = e_{gh}$.

Exercise 3 (The four module categories over Vec_{S_3}). You may use that $H^2(H, \mathbb{C}^\times) = 0$ for every subgroup H of S_3.

(i) List the subgroups of S_3 up to conjugacy, and the module categories of Theorem 11.11, with their numbers of simple objects.

(ii) For $H = \mathbb{Z}_3$, write the action of a transposition and of a 3-cycle on the simple objects.

(iii) Over $\mathrm{Rep}(S_3)$ the same pairs give $\mathrm{Rep}(H)$ (Lecture 11, cited). How many simple objects does each have?

Hint. (i) The simple objects of $\mathrm{Mod}(\mathbb{C}[H])$ are the cosets gH. (iii) Count irreducible representations.

Answer. (i) 1, $\langle(12)\rangle$ (three conjugates), $\mathbb{Z}_3$, S_3; ranks $6, 3, 2, 1$. (ii) The two cosets of $\mathbb{Z}_3$: transpositions swap them, 3-cycles fix them. (iii) $1, 2, 3, 3$.

Exercise 4 (NIM-reps; Fibonacci). (i) Show that the NIM-rep of a semisimple module category satisfies $n_1 = 1$ and $n_a n_b = \sum_c N_{ab}^c \, n_c$; you may use that $n_{a^*} = n_a^T$ (duality).

(ii) For Fibonacci, show that a symmetric matrix n of size $r \geq 1$ with non-negative integer entries and $n^2 = 1 + n$ that is indecomposable (no nonempty proper set of simples closed under n) is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ up to relabelling. Which module category has this NIM-rep?

(iii) Let $\mathcal{M}$ be a nonzero indecomposable semisimple module category over Fibonacci, with finitely many simple objects. Show that its NIM-rep is indecomposable. Then take a simple M with $(n_\tau)_{MM} = 0$, and use Theorems 11.8 and 11.10 to show that $\mathcal{M}$ is equivalent to the regular module C.

Hint. (i) Count $\dim \mathrm{Hom}(a \otimes (b \otimes M_j), M_i)$ two ways. (ii) Compare the diagonal entries of n^2 with those of $1 + n$, then the off-diagonal ones. (iii) n_τ is symmetric. An algebra structure on the object 1 is a pair of scalars.

Answer. (i) $(a \otimes b) \otimes M_j \cong a \otimes (b \otimes M_j)$ and both sides decompose into the M_i. (ii) Each row has $n_{ii} \in \{0, 1\}$ and one other 1; a pair i, j with $n_{ii} + n_{jj} = 1$; the regular module. (iii) A closed set and its complement would split $\mathcal{M}$; $[M, M] = 1$, an algebra isomorphic to 1, so $\mathcal{M} \simeq \mathrm{Mod}(1) = C$.

Exercise 5 ($\mathrm{SU}(2)_4$: A_5 and D_4). For $\mathrm{SU}(2)_4$, a NIM-rep is determined by $n = n_{1/2}$ through $n_0 = 1$ and $n_{j+1/2} = n\, n_j - n_{j-1/2}$ (Sheet 5, Exercise 2).

(i) The Dynkin diagram A_5 is the path on the vertices $1, 2, 3, 4, 5$, and D_4 is the graph with a centre 2 joined to three ends $1, 3, 4$. Draw them, and let n be the adjacency matrix of either. Show that the n_j, $j = 0, \ldots, 2$, are non-negative integer matrices and that $n_{5/2} = 0$, and deduce that the n_j form a NIM-rep.

(ii) Show that the eigenvalues of n are $2\cos(m\pi/6)$ for $m = 1, \ldots, 5$ (A_5) and $m = 1, 3, 3, 5$ (D_4), and explain why $n_{5/2} = 0$ follows from this.

Hint. (i) Write $n_1, \ldots, n_{5/2}$ as polynomials in n. For the last part, Sheet 5, Exercise 2(ii) writes every $[j]$ as a polynomial in $[\frac{1}{2}]$. (ii) $n_j = U_{2j}(n)$, and n is symmetric.

Answer. (i) A_5: $n_1 = n^2 - 1$, n_2 the reversal $i \mapsto 6 - i$, $n_{3/2} = n\, n_2$; D_4: $n_1 = n^2 - 1$ (centre entry 2), $n_{3/2} = n$, $n_2 = 1$. In both $n_{5/2} = n\, n_2 - n_{3/2} = 0$, and $x \mapsto n$ is a ring map from $K(\mathrm{SU}(2)_4) \cong \mathbb{Z}[x]/(U_5)$. (ii) $n_{5/2} = U_5(n)$ vanishes on each eigenvalue $2\cos(m\pi/6)$, since $\sin(6 \cdot m\pi/6) = 0$.

Exercise 6 ⋆. *Ising has only the regular module.* A NIM-rep of the Ising ring is a pair of symmetric $r \times r$ ($r \geq 1$) non-negative integer matrices n_σ, n_ψ with $n_\psi^2 = 1$, $n_\sigma n_\psi = n_\sigma$ and $n_\sigma^2 = 1 + n_\psi$.

(i) Show that n_ψ is the permutation matrix of an involution π of the simple objects, and that each row of n_σ is unchanged by π.

(ii) Show that if π moves i, then row i of n_σ has a single 1, at a point j that π fixes, and row j is $e_i + e_{\pi(i)}$.

(iii) Show that π cannot fix every simple object, and conclude that every indecomposable NIM-rep is the regular one, with three simple objects.

(iv) Conclude, as in Exercise 4(iii), that every nonzero indecomposable semisimple module category over Ising, with finitely many simple objects, is equivalent to the regular one.

Hint. (i) A non-negative integer matrix whose inverse is non-negative too. (ii) Compare the diagonal of n_σ^2 with that of $1 + n_\psi$. (iii) If $\pi = \mathrm{id}$, look at the off-diagonal entries of $n_\sigma^2 = 2$. (iv) Which simple has zero diagonal entries in both n_σ and n_ψ?

Answer. (i) $n_\psi^{-1} = n_\psi \geq 0$ forces a permutation; $n_\sigma n_\psi = n_\sigma$ permutes the columns. (ii) The row has square-sum 1 (one 1) at a fixed point j, whose row has square-sum 2 and contains $i, \pi(i)$. (iii) For $\pi = \mathrm{id}$, $n_\sigma^2 = 2$ fails off the diagonal; so the component of a moved point is $\{i, \pi(i), j\}$, the regular NIM-rep. (iv) At a point i moved by π, $(n_\sigma)_{ii} = (n_\psi)_{ii} = 0$, so $[M_i, M_i] = 1$ and $\mathcal{M} \simeq \mathrm{Mod}(1) = C$.